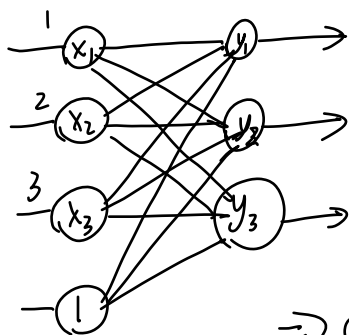


Affine space and its generation 1: Vector space and its generation.



w_{11} : weight from x_1 to y_1 , b_1 : weight from 1 to y_1

$$y_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + b_1$$

$$y_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 + b_2$$

$$y_3 = w_{13}x_1 + w_{23}x_2 + w_{33}x_3 + b_3$$

$$\Rightarrow \text{compact form: } \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \underbrace{\begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \\ w_{13} & w_{23} & w_{33} \end{bmatrix}}_W \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_X + \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_b$$

$$\Rightarrow \boxed{\vec{y} = W\vec{x} + \vec{b}} \rightarrow \text{affine mapping}$$

linear mapping $\rightarrow$ Vector space

1. Group

Consider a set G and an operation: $\otimes: G \times G \rightarrow G$ defined over G .
 Then $G: (G, \otimes)$ is called a group if the following hold:

- ① $\forall x, y \in G, x \otimes y \in G$: The set is closed w.r.t. the operation
- ② $\forall x, y, z \in G, (x \otimes y) \otimes z = x \otimes (y \otimes z)$ Associativity
- ③ $\exists e \in G$ s.t. $\forall x \in G, x \otimes e = e \otimes x = x$ Identity element.
- ④ $\forall x \in G, \exists y \in G$ such that $x \otimes y = y \otimes x = e$. We denote by x^{-1} the inverse element of x . Every element has its inverse in its group

Specifically, G is an Abelian group if.

- ① G is a group: ② $\forall x, y \in G, x \otimes y = y \otimes x$ Commutativity

① Example $(\mathbb{R} \setminus \{0\}, \cdot)$ is an Abelian group

proof: ① $\forall a, b \in \mathbb{R} \setminus \{0\}, a \cdot b \in \mathbb{R} \setminus \{0\}$

$$\text{② } \forall a, b, c \in \mathbb{R} \setminus \{0\}, (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

$$\text{③ consider } 1 \in \mathbb{R} \setminus \{0\}, \forall a \in \mathbb{R} \setminus \{0\}, a \cdot 1 = 1 \cdot a = a.$$

$$\text{④ } \forall a \in \mathbb{R} \setminus \{0\}, \text{ we have } a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$$

$$\text{⑤ } \forall a, b \in \mathbb{R} \setminus \{0\}, a \cdot b = b \cdot a$$

Exercise: $(\mathbb{R}, +)$ is an Abelian Group.

① $\forall a, b \in \mathbb{R}, a+b \in \mathbb{R}$

② $\forall a, b, c \in \mathbb{R}, (a+b)+c = a+(b+c)$

③ consider $0 \in \mathbb{R}, \forall a \in \mathbb{R}, a+0=0+a=a$.

④ $\forall a \in \mathbb{R},$ we have $a+(-a) = -a+a = 0$

⑤ $\forall a, b \in \mathbb{R}, a+b = b+a$

Example: $(\mathbb{R}, \cdot)$ is not a group

④ for $0 \in \mathbb{R}, \nexists a \in \mathbb{R}, \text{ s.t. } a \cdot 0 = 1 \quad \checkmark$

2. Vector space

A real-valued vector space $V := (V, +, \cdot)$ is a set with two operations:

1) $+: V \times V \rightarrow V$
addition of vectors

2) $\cdot: \mathbb{R} \times V \rightarrow V$
multiplication of vector by scalars

where 1) $(V, +)$ is an Abelian group

2) Distributivity (over $\mathbb{R}$) holds.

① $\forall \lambda \in \mathbb{R}, \forall \vec{x}, \vec{y} \in V: \lambda(\vec{x} + \vec{y}) = \lambda \cdot \vec{x} + \lambda \cdot \vec{y}$

② $\forall \lambda, \psi \in \mathbb{R}, \forall \vec{x} \in V: (\lambda + \psi) \cdot \vec{x} = \lambda \cdot \vec{x} + \psi \cdot \vec{x}$

3) $\forall \lambda, \psi \in \mathbb{R}, \forall \vec{x} \in V, \lambda \cdot (\psi \cdot \vec{x}) = (\lambda \psi) \cdot \vec{x}$

4) $\forall \vec{x} \in V: 1 \cdot \vec{x} = \vec{x}$

space: a set with special structure among elements
operation

3. Vector subspace

Let ① A vector space $V = (V, +, \cdot)$; ② $U \subset V$ and $U \neq \emptyset$

Then, $U := (U, +, \cdot)$ is called a vector subspace of V , if

① U is a vector space with the operation $+$ and $\cdot$ restricted to $U \times U$ and $\mathbb{R} \times U$, respectively. $\Rightarrow$ Denote by $U \subseteq V$

3. 1) Linear combination.

Consider a vector space $V := (V, +, \cdot)$, and a finite number of vectors $\vec{x}_1, \dots, \vec{x}_k \in V$. Then, every $\vec{v} \in V$ of the form

$$\vec{v} = \lambda_1 \vec{x}_1 + \lambda_2 \vec{x}_2 + \dots + \lambda_k \vec{x}_k = \sum_{i=1}^k \lambda_i \vec{x}_i \in V$$

with $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ is a linear combination of vectors $\vec{x}_1, \dots, \vec{x}_k$.

↓ 2) Linear dependency

Consider a vector space V , $k > 0$, vector $\vec{x}_1, \dots, \vec{x}_k \in V$, and

$$\sum_{i=1}^k \lambda_i \vec{x}_i = \vec{0} \quad \text{for some } \lambda_i \in \mathbb{R}, i \in [1, k].$$

zero vector

We say $\vec{x}_1, \dots, \vec{x}_k$ are -

- ① Linearly independent: only trivial solution exists: i.e. $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$
- ② Linearly dependent: nontrivial linear combination exists, i.e. $\lambda_i \neq 0$ for some $i \in [1, k]$

3) Spanning / Generating sets

Consider a vector space $V := (V, +, \cdot)$ and a set

$A := \{\vec{x}_1, \dots, \vec{x}_k\} \subseteq V$, if $\forall \vec{v} \in V, \exists \lambda_1, \dots, \lambda_k \in \mathbb{R}$, s.t.

$$\vec{v} = \sum_{i=1}^k \lambda_i \vec{x}_i \quad \rightarrow \quad \lambda_i: \text{coordinate}$$

$\Rightarrow A$ is called a spanning / generating set of V , denoted by $\text{Span}(A) = V$ or A spans V .

4) minimal generating set (a.k.a. basis)

A is the minimal generating set of V , if

- ① A is a generating set of V : ② $\nexists A' \subset A$, s.t. A' spans V

Take away

- ① basis: a minimal set of linear independent vectors that spans V
- ② in general, vectors in a generating set could be linearly dependent
- ③ Dimension: the number of basis vectors; denoted by $\dim(V)$

Affine space and its generation: linear mapping affine mapping

Another way to viewing $\underline{W}\vec{x}$, let $\vec{x} := [x_1 \ x_2 \ x_3]^T$

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_{11}x_1 + w_{12}x_2 + w_{13}x_3 \\ w_{21}x_1 + w_{22}x_2 + w_{23}x_3 \\ w_{31}x_1 + w_{32}x_2 + w_{33}x_3 \end{bmatrix} = x_1 \cdot \begin{bmatrix} w_{11} \\ w_{21} \\ w_{31} \end{bmatrix} + x_2 \cdot \begin{bmatrix} w_{12} \\ w_{22} \\ w_{32} \end{bmatrix} + x_3 \cdot \begin{bmatrix} w_{13} \\ w_{23} \\ w_{33} \end{bmatrix}$$

column space of W

$\Rightarrow \tilde{W}\vec{x}$ is a vector within the span of column vectors of $\tilde{W}$

$\Rightarrow \tilde{W}\vec{x}$ is a linear transformation of $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ from its own space (denoted by V) to the column space of $\tilde{W} \Rightarrow$ a mapping from V to W

1. Linear mapping

Def (Linear mapping) Given two vector spaces V, W ,

a mapping $\phi: V \rightarrow W$ is a linear mapping if

$$\forall \vec{x}, \vec{y} \in V, \forall \lambda, \psi \in \mathbb{R}, \phi(\lambda\vec{x} + \psi\vec{y}) = \lambda\phi(\vec{x}) + \psi\phi(\vec{y})$$

Specifically: ① $\lambda = \psi = 1, \phi(x+y) = \phi(x) + \phi(y)$

$$\text{② } \psi = 0, \lambda = 1, \phi(\lambda\vec{x}) = \lambda\phi(\vec{x})$$

$\Rightarrow$ linear mapping preserves the structure of vector space

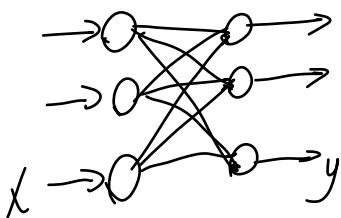
$\Rightarrow$ the order between linear mapping and operations of the vector space can be exchange.

given $\vec{x} \in V$, $\tilde{W}\vec{x}$ maps $\vec{x}$ to space W , let $\vec{x} \in \mathbb{R}^n$
 $\tilde{W}\vec{x} \in \mathbb{R}^m$ we denote the number of linearly independent columns of a matrix W by $\text{rank}(W)$, or the rank of $\tilde{W}$

important property: for $\tilde{W} \in \mathbb{R}^{m \times n}$, $\text{rank}(\tilde{W}) = \min\{m, n\}$, then we say $\tilde{W}$ has full rank.

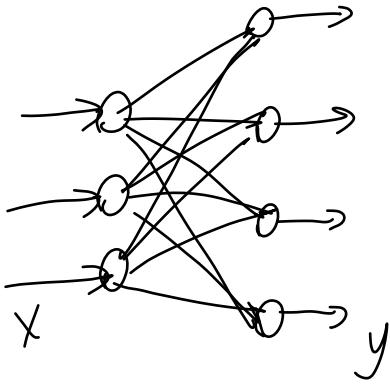
Three situation (consider $\tilde{W}$ with full rank).

① $\text{rank}(\tilde{W}) = n = m$, space V and W has the same dimension.
we are representing a "same" vector using different coordinate.



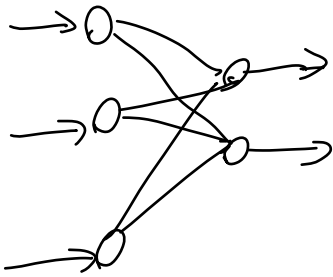
y (output) has a different interpretation of x (input)

② $\text{rank}(\tilde{w}) = n < m$, meaning that W has a higher dimension than V , we are mapping the input data to a higher dimensional space (a.k.a upsampling).



handling data in a lower dimension
in a higher dimension

③ $\text{rank}(\tilde{w}) = m < n$, meaning that W has a lower dimension than V , we are mapping the input data to a lower dimensional space (a.k.a. downsampling).



dimensionality reduction.

→ handling "the most important" perspective of some high dimensional data.

⇒ what if $\tilde{w}$ does not have full rank?

⇒ some connections are not necessary,

⇒ more property of rank: session 2.6.2

Exercise: basis change.

Consider vector space V and W , and ordered bases

$$B = (\vec{b}_1, \dots, \vec{b}_n), \tilde{B} = (\vec{b}'_1, \dots, \vec{b}'_n) \text{ of } V$$

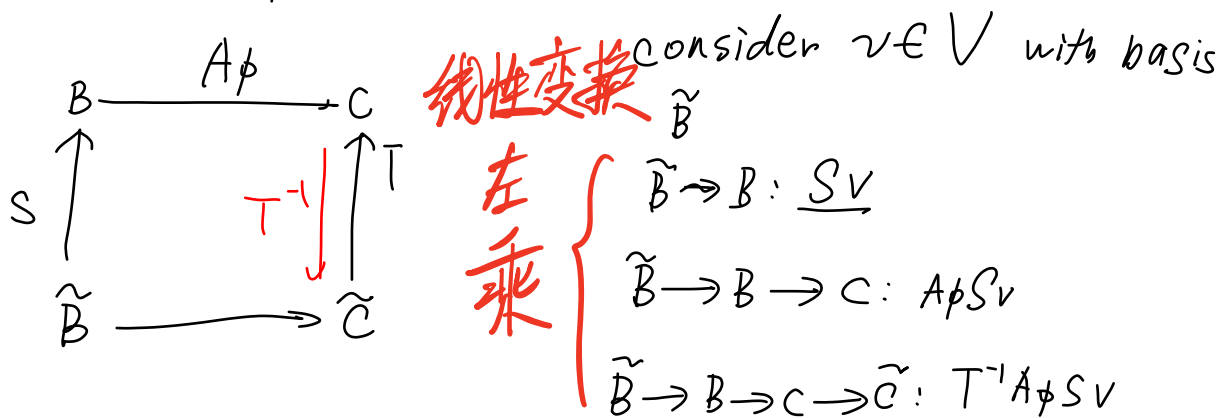
$$C = (\vec{c}_1, \dots, \vec{c}_m), \tilde{C} = (\vec{c}'_1, \dots, \vec{c}'_m) \text{ of } W$$

Let $\underline{A}$, $\underline{S}$, $\underline{T}$ be transformation from B to C , from $\tilde{B}$ to B and from $\tilde{C}$ to C , respectively.

$\Rightarrow$ transformation matrix $\tilde{B}$ to $\tilde{C}$

$$\tilde{A}\phi = ?$$

Answer.



$$\Rightarrow \tilde{A}\phi = T^{-1}A\phi S$$

Remark 1: Matrices $A\phi$ and $\tilde{A}\phi$ are **equivalent** if there exist regular matrix (i.e. matrix with full rank) $S \in \mathbb{R}^{n \times n}$ and $T \in \mathbb{R}^{m \times m}$ such that $\tilde{A}\phi = T^{-1}A\phi S$

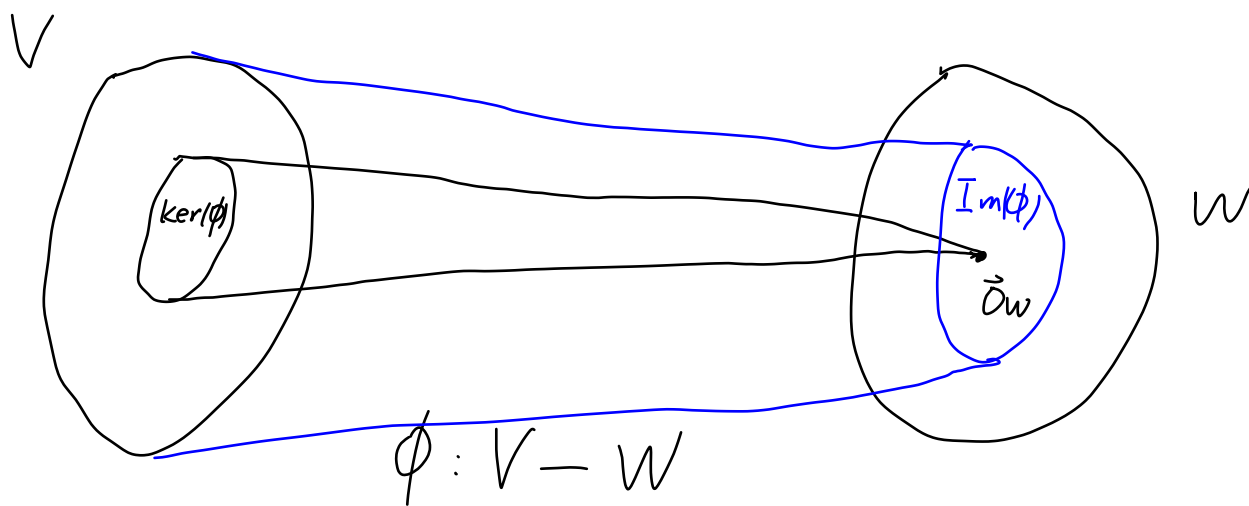
Remark 2: Matrices $A\phi$ and $\tilde{A}\phi$ are **similar** if there exist a regular matrix $S \in \mathbb{R}^{n \times n}$, s.t. $\tilde{A}\phi = S^{-1}A\phi S$
 (Transformation matrix from $\tilde{B}$ to B , and $\tilde{C}$ to C , are the same)

2. Image and kernel

For $\phi: V \rightarrow W$ (not restricting to linear mapping), we define.

$$\text{Kernel: } \ker(\phi) := \phi^{-1}(\vec{0}_W) = \{ \vec{v} \in V \mid \phi(\vec{v}) = \vec{0}_W \}$$

$$\text{Image: } \text{image}(\phi) = \{ \vec{w} \in W \mid \exists \vec{v} \in V, \text{ s.t. } \phi(\vec{v}) = \vec{w} \}$$



further more: if ϕ is a linear transformation with transformation matrix A

kernel: $\ker(\phi) := \{ \vec{v} \in V \mid A\vec{v} = \vec{0}_W \}$ Null space

$\Rightarrow$ Capture all possible linear combination of elements in V that produce $\vec{0}_W$

$\text{Image}(\phi) = \{ \vec{w} \in W \mid \exists \vec{v} \in V, \text{ s.t. } A\vec{v} = \vec{w} \}$

$\Rightarrow$ span of column vectors of A : column space

more properties: session 2.7.3 for linear mapping

3. Affine Space

Def. Let V be a vector space, $\vec{x}_0 \in V$, and $U \subseteq V$ be a subspace. Then

$$\begin{aligned} \underline{L} &= \vec{x}_0 + U := \{ \vec{x}_0 + \vec{u} \mid \vec{u} \in U \} \\ &= \{ \vec{v} \in V \mid \exists \vec{u} \in U : \vec{v} = \vec{x}_0 + \vec{u} \} \subseteq V \end{aligned}$$

is called affine subspace (or linear manifold) of V

Exercise: prove: if $\vec{x}_0 \notin U$, $\vec{0} \notin L$
 $\vec{x}_0$: support point U : direction space

proof: Suppose $\vec{x}_0 \notin U$, but $\vec{0} \in L$, then $\exists \vec{u} \in U$, s.t. $\vec{x}_0 + \vec{u} = \vec{0}$

$$\Rightarrow -\vec{x}_0 = \vec{u} \in U$$

since U is a vector subspace, the $(U, +)$ is an Abelian group

therefore, $-\vec{x}_0 \in U$, $\exists \vec{a} \in U$, s.t. $-\vec{x}_0 + \vec{a} = \vec{0}$

$\Rightarrow \vec{a} = \vec{x}_0 \in U$, which is contradictory to $\vec{x}_0 \notin U$ ✓

Lemma: if $\vec{x}_0 \notin U$, L is not a vector subspace.

4. Affine mapping.

Given two vector spaces V, W , a linear mapping $\phi: V \rightarrow W$, and $\vec{a} \in W$, the mapping $\phi': V \rightarrow W$ with

$$\phi'(\vec{x}) = \vec{a} + \phi(\vec{x})$$

translation vector

is called an affine mapping from V to W

Sec 2-4 ~ 2-9